Computers in the Past, Present and Future

Computers have come a long way from room-sized machines crunching basic equations to sleek, portable powerhouses. Early processors could only handle simple instructions and ran at speeds of only a few kilohertz, while today's multi-core CPUs can handle billions of complex operations in a single second.

System memory has come a long way, from a few kilobytes of crude memory to dozens of gigabytes of lightning-fast RAM that enables you to multitask with ease.

Meanwhile, storage has gone from huge, clunky disks that stored a few megabytes to tiny solid-state drives that can hold terabytes. These rapid leaps in processing, memory and storage have combined to make modern computing exponentially faster, smaller and more accessible.

Generations of computers

Comparison Table Template

Feature	First Generation	Second Generation	Third Generation	Fourth Generation	Fifth Generation
Technology	Vacuum Tubes	Transistors	Integrated Circuits	Microprocessors	AI, Quantum Computing, Nanotech
Size	Very large	Smaller	More compact	Personal computers	Varies
Speed and Storage	Slow, kilobytes	Faster, megabytes	Significantly faster, more memory	Very fast, gigabytes to terabytes	Extremely fast, advanced storage
Examples	ENIAC, UNIVAC	IBM 1401, IBM 7090	IBM 360, PDP-8	Intel 4004, Apple II, IBM PC	IBM Watson, Google DeepMind
Programming	Machine language	Assembly, early high-level	High-level (COBOL, FORTRAN)	High-level, object-oriented	AI languages, modern high-level
Characteristics	High power consumption, unreliable	More reliable, less power	Increased reliability, OS support	Greatly increased speed, GUIs	AI, machine learning, quantum computing

In the past

First Generation [1940 – 1956]

These room-sized computers used vast amounts of electricity, generated enormous heat, and routinely failed. They had limited memory and were entirely dependent on monotonous machine code to do simple calculations. One characteristic was their dependence on physical punched cards and paper tape for all data input and output.

Second Generation (Transistors) [1956 – 1963]:

The second generation replaced glass tubes with solid-state transistors, which used semiconductor materials such as silicon to control electrical signals, rather than a fragile vacuum. With just a tiny voltage applied, the transistor could switch or amplify electrical currents instantly, with almost no heat or wear and tear. This breakthrough resulted in computers that were dramatically smaller, faster, and more reliable. It also paved the way for assembly code and the first high-level programming languages including FORTRAN and COBOL. During this period, magnetic core memory was introduced, as were magnetic tape reels, which provided a reusable, and far more compact, form of data storage.

Third Generation (Integrated Circuits) [1964 – 1971]:

In the third generation, the computing process was further developed through the printing of several hundred to thousands of tiny transistors on the same silicon chip known as an integrated circuit (IC). Through doing away with wiring different parts by hand, signal transmission distances were shortened dramatically, leading to higher execution speeds. The computers became more compact, and the first multitasking operating systems came into play as well. Notably, the use of keyboards and computer screens meant that the machine could be operated through direct user interaction as opposed to output cards.

In the Present

Fourth Generation (Microprocessors) [1971 – Present]:

The fourth generation marked the appearance of the microprocessor, which combined the computing capabilities of an entire computer, millions, then billions, of tiny transistors on a single microchip. Computing became highly powerful and highly affordable because the integration of all processing functions in one silicon chip increased computing abilities while reducing costs; thus, personal computers, laptops, and smartphones were born. The fourth generation was characterized by terabytes of solid-state memory and Internet connection across the globe. The Graphical User Interface was its hallmark.

Fifth Generation – Present [2010s – Present]:

The present-day computing era is that of the fifth generation, which makes use of massively parallel processors (GPUs and neural processing units) to handle huge volumes of data streams at once instead of handling them one after another. The computers, by emulating neural networks, handle human languages, recognize images, and learn from large volumes of data. These computers constitute today's artificial intelligence algorithms, cloud supercomputers, and automated software. One unique characteristic of this era is natural language processing.

In the Future

Beyond the Present:

Future technologies go beyond the traditional use of silicon microchips, employing quantum mechanics and nanotechnology at a molecular scale. Quantum computers do not make use of traditional bits, which either have to be 0 or 1; qubits employed by quantum computers can exist in various states at the same time and thus solve highly complex mathematical and molecular computations all at once. When coupled with neuromorphic computers, future computers try to solve problems like the creation of new drugs, climate simulations, and even true artificial general intelligence, problems that even the fastest supercomputers will take thousands of years to compute.